Tectonophysics, 60 (1979) 189—205

© Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

PRELIMINARY MESOZOIC PALAEOMAGNETIC RESULTS FROM ISRAEL AND INFERENCES FOR A MICROPLATE STRUCTURE IN THE LEBANON

R. FREUND and D.H. TARLING

Department of Geology, The Hebrew University, Jerusalem (Israel)
Department of Geophysics and Planetary Physics, The University, Newcastle-upon-Tyne (Great Britain)

(Received April 7, 1978; revised version accepted October 26, 1978)

ABSTRACT

Freund, R. and Tarling, D.H., 1979. Preliminary Mesozoic palaeomagnetic results from Israel and inferences for a microplate structure in the Lebanon. Tectonophysics, 60: 189—205.

Preliminary results from a few Triassic, Jurassic and Cretaceous rocks indicate that Israel has behaved as part of the African plate during this period. Earlier palaeomagnetic results from the Lebanon, previously explained in terms of a separate Levantine plate, can be better explained in terms of differential motion between fault blocks in response to motions along the Dead Sea Fault. The available evidence allows determination of the net motions of specific fault blocks and is consistent with independent palaeogeographic indicators of the pretectonic relationship between these blocks.

INTRODUCTION AND SAMPLING

The first palaeomagnetic results from the Lebanon were obtained by Van Dongen et al. (1967) who determined a Jurassic pole position near the Equator, some 70° anticlockwise from contemporaneous African poles, and a Cretaceous pole near Florida, some 40° anticlockwise from African Cretaceous poles. These results also contrasted with later results from Israel (Helsley and Nur, 1970) who determined pole positions close to those of Africa. The area from which the apparently anomalous Lebanese data were obtained was resampled and confirmed by Gregor et al. (1974). They suggested that Lebanon may have acted as a separate microplate with its southern boundary just north of Mt. Carmel in Israel. A further possibility is that Lebanon consists of a series of faulted blocks which have rotated individually in response to the 105 km post-Cretaceous shear along the Dead Sea Rift (Freund et al., 1970). In order to further elucidate this problem, a preliminary survey was carried out in Israel, mainly to establish whether the exposed sedimentary

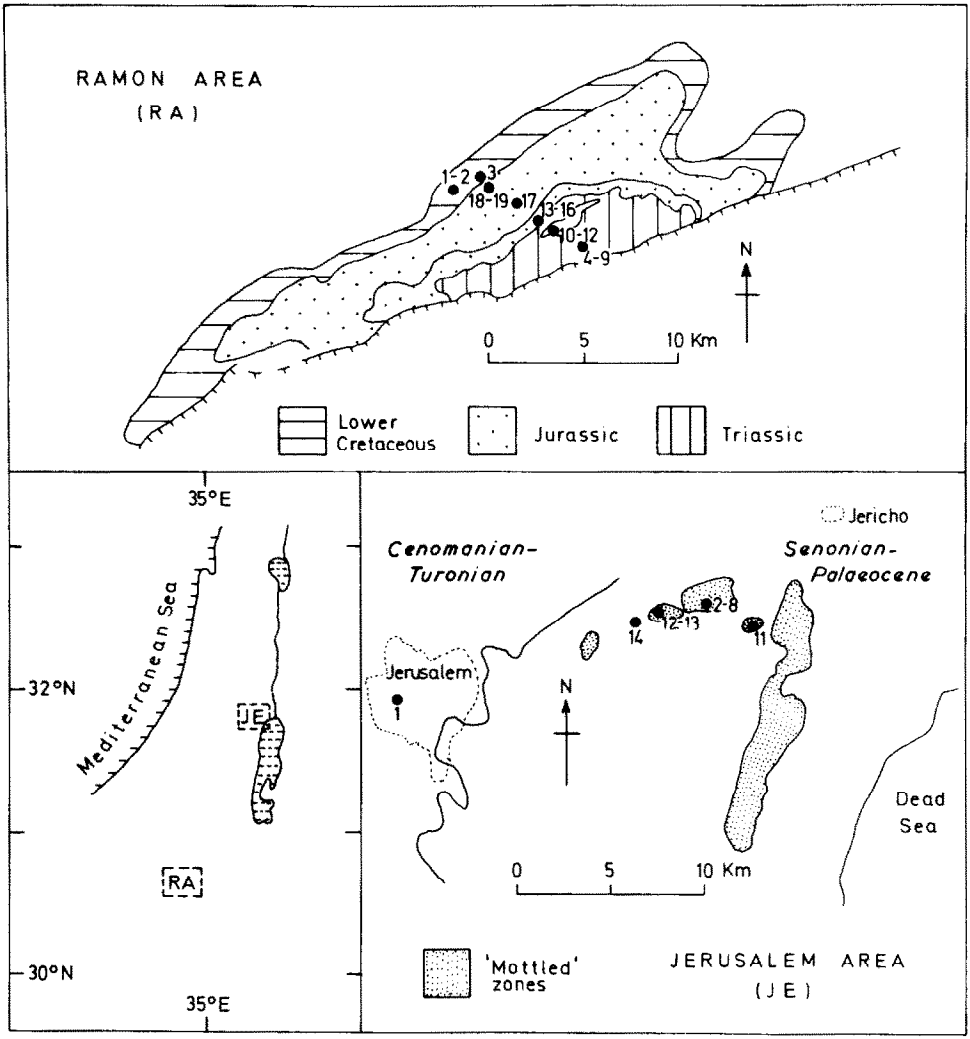


Fig. 1. Sampling sites. Two areas were sampled; the Ramon (*RA*) and Jerusalem (*JE*) regions. Of the *JE* sites, only those away from the autometamorphosed zone are discussed here.

rocks were suitable for further palaeomagnetic study.

Sites were drilled in the Ramon anticline (RA) and the Jerusalem—Jericho area (JE), with a minimum of six separately sun-compass oriented samples being collected at each site (Fig. 1). The Ramon anticline strikes ENE and exposes Triassic formations at its core, with Jurassic and Lower Cretaceous rocks exposed on its flanks. The folding is thus post-Lower Cretaceous and most likely associated with Tertiary compression along the western side of the Dead Sea Rift. A total of 19 sites was collected on both flanks of the

anticline and these included extrusive and intrusive rocks as well as sediments. Within Jerusalem, one site was sampled in Cenomanian limestones (JE1) and near Jericho two sites were collected (JE8 and JE10). Other sites near Jericho were collected as part of a separate investigation into the origin of "mottled" zones, attributed to autometamorphism, within these limestones (Freund and Tarling, in prep.) but the sites used here were well away from any possible metamorphic effects.

PALAEOMAGNETIC RESULTS

One specimen was cut from each oriented sample. The remanence of each sample was then measured (Table I) and one pilot specimen from each site was subjected to incremental partial demagnetization in order to determine the optimum field for isolating the most stable direction of remanence (Table II). Igneous rocks were treated in alternating magnetic fields and sediments were thermally demagnetised. All remaining specimens were then partially demagnetised in the optimum field or temperature, indicated by the pilot specimen behaviour (Table II).

Triassic sediments (Fig. 2)

Four of the six sites had internal scatters less than 20° before partial demagnetisation and these were all grouped close to the present geomagnetic field. The grouping of the site means was better before correction for tectonic tilt, suggesting that a significant component of their natural remanence had been acquired after formation of the Ramon anticline. The two scattered sites (RA6 and RA9) were both characterised by weak intensity, although site RA9 also included some strongly magnetised samples. Thermal demagnetisation of the pilot specimen yielded variable results. Specimens RA4.1, 6.2 and 9.1 each changed direction by substantial amounts, but then became consistently centered either easterly (RA4.1) or southerly (RA6.2, 9.1) directions. Specimen RA7.3 remained close to the present geomagnetic field direction until 150°C, but became too weak to measure by 250°C and specimen RA5.1 similarly became unmeasurable above 200°C although initially had a direction similar to that of the laccolith (RA10—12). On the basis of this behaviour, all remaining samples were partially demagnetised at 300°C. Sites RA5, 7 and 8 all became widely scattered, but the other three sites (RA4, 6 and 9) were tightly grouped, but dissimilar to each other. Closer examination of the higher temperature behaviour of the pilot specimens does not reveal any obvious multicomponent magnetisation and the within-site distribution of sample directions does not indicate streaking. These three site mean directions, however, are too scattered to be consistent with secular variation and, at this stage, it is felt that they cannot be regarded as reliable estimates of the Triassic geomagnetic field.

TABLE I

Site mean directions

Site	Initial			Demagnetised (300°C or 15 · 10^{-3} Wb m^{-2})					
	decl. (degr.)	inc.l (degr.)	c.s.d	decl. (degr.)	incl. (degr.)	c.s.d.	lat. (degr.)	long. (degr.)	A_{95} (degr.)
Upper Cretaceous (JE)									
Deir Yasini limestone (Giv'at Ram Campus)									
1	337	37	15	336 *	17 *	18	58N	84W	14
Senonian (JE)									
8	(27	+74)	58	189	—40	19	77.8S	8W	14
10	(19	+50)	78	147	+44	10	24.2S	68E	7
Lower Cretaceous (RA)									
Basalt									
3	(82	—9)	27	(57	7)	36	—	—	32
Marine intercalation									
1	(76	36)	44	171	—43	3	80S	91E	3
2	(299	30)	60	176	—43	4	84S	67E	4
Jurassic—Cretaceous									
Essexite laccolith									
10	342	23	4	329	—4	6	46N	87W	4

11	338	16	4	323	1	2	44N	88W	2
12	341	19	5	335	9	6	55N	98W	3
Mid Jurassic (RA)									
Inmar Fm									
17	(353	38)	34	352	31	18	75N	115W	13
Daya (Mahmal) Fm.									
18	358	39	2	(3	29)	51	—	—	45
19	(156	−57)	50	165	−51	4	77S	134E	4
Lower Jurassic (RA)									
Ardon Fm									
15	357	33	4	357	32	4	76N	131W	3
14	4	28	16	7	30	17	74N	169W	13
16	343	21	11	335	16	10	57N	94W	7
Triassic (RA)									
Saharonim Fm.									
6	(62	39)	31	170	19	9	48S	50E	5
4	49	36	12	74	34	12	24N	116E	10
7	7	32	5	(26	87)	42	—	—	60
5	358	34	13	(348	58)	41	—	—	48
9	(299	−3)	66	193	−32	9	72S	10W	6
8	4	39	5	(201	75)	68	—	—	—

* JE1 demagnetised at. 250°C.

TABLE II

Most stable directions of pilot samples

Specimen	Decl. (degr.)	Incl. (degr.)	A_{95} (degr.)	S.I.	Specimen	Decl. (degr.)	Incl. (degr.)	A_{95} (degr.)	S.I.
Upper Cretaceous (JE)					Mid Jurassic (RA)				
1.6	70.5	67.9	6	2.4	17.2	351.6	36.4	2	7.9
8.1	179.9	−51.0	7	1.9	18.1	59.9	−72.1	30	0.5
					19.1	166.1	−44.9	1	14.3
Lower Cretaceous (RA)									
3.1 *	26.8	8.1	13	1.0	Lower Jurassic (RA)				
1.1	168.7	−41.3	3	7.2	14.1	350.3	36.9	2	5.3
2.2	173.4	−42.4	2	6.5	15.1	352.6	32.2	2	6.9
					16.1	324.1	6.7	4	3.6
Cretaceous—Jurassic laccolith (RA)									
10.1 *	326.4	11.3	6	2.4	Triassic (RA)				
11.1 *	323.8	3.6	2	6.2	6.1	179.8	5.6	5	2.9
12.1 *	331.2	2.7	2	5.3	4.1	73.5	31.5	2	8.1
					7.3	359.1	36.9	8	1.6
					5.1	322.1	34.7	11	1.2
					9.1	198.6	−39.0	3	5.1

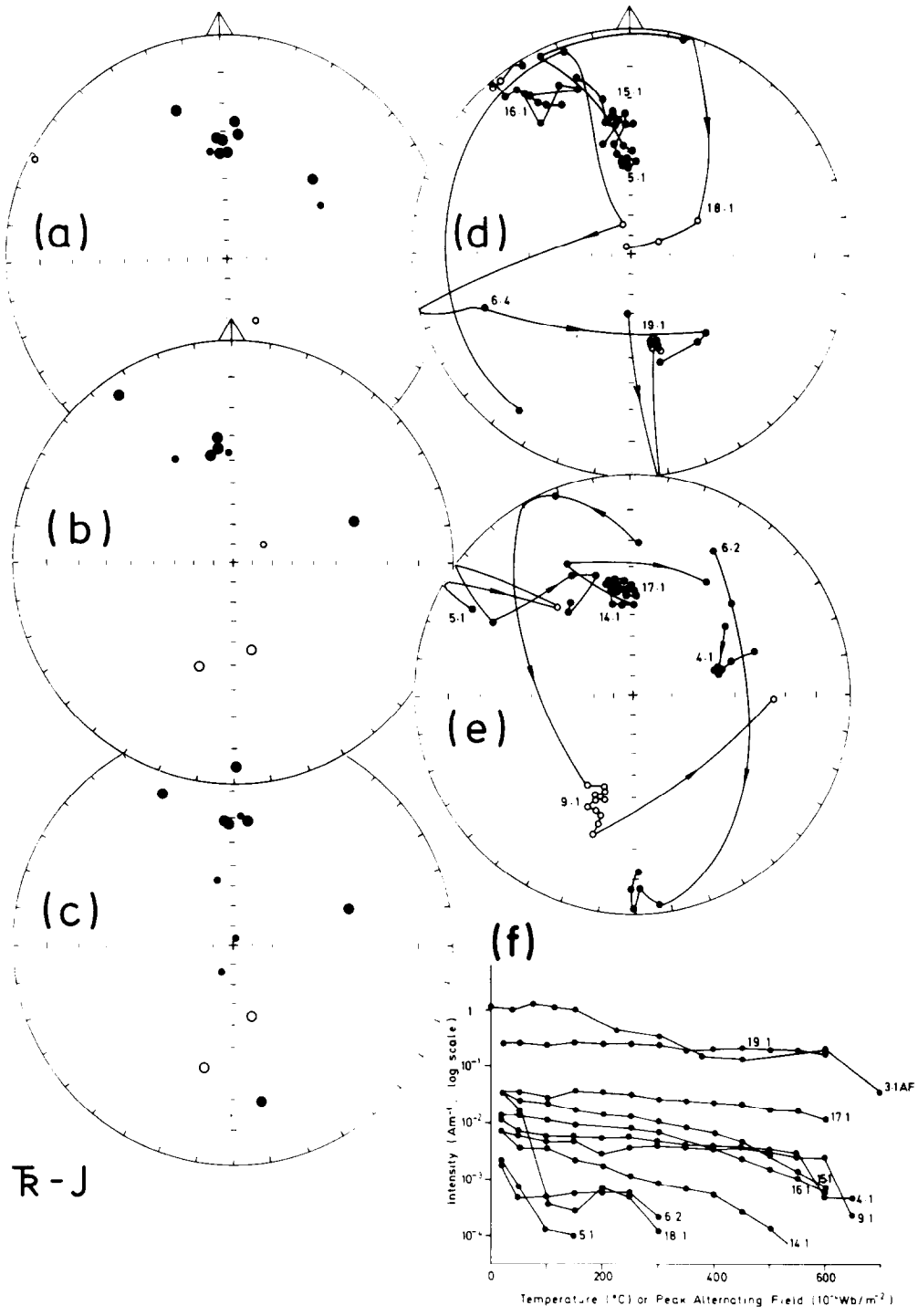
All specimens were subjected to incremental thermal demagnetisation, except those marked * which were igneous rocks which were subjected to alternating magnetic fields. Directions are all corrected for tectonic tilt. S.I. = stability index (Tarling and Symons, 1967) in which S.I. $\geqslant 2.5$ is considered very stable, and S.I. < 1.0 is completely unstable. c.s.d. = circular standard deviation.

Jurassic sediments (Fig. 2)

Four of the six sites have initial circular standard deviations less than 20°, of which three are close to the present geomagnetic field direction. On incremental thermal demagnetisation, the intensities of strongly magnetised, Middle Jurassic pilot specimens (RA 17.1, RA 19.1) showed only a very

Fig. 2. Triassic—Jurassic directions and intensities of magnetisation. All directions are plotted after correction for tectonic tilt, on stereographic projections with the circumference as horizontal. Solid symbols correspond to downward (positive inclinations of remanence. Data for igneous rocks are shown as triangles and these were subjected to alternating magnetic field treatment while the circular symbols correspond to sedimentary rocks which were thermal demagnetised.

Larger symbols on (a) and (c) correspond to site mean directions with a circular standard deviation of less than 20° in which (a) is for sites prior to demagnetisation and (c) to site directions after thermal or alternating magnetic field partial demagnetisation. In (b) the directions are for the most stable direction of magnetisation (Tarling and Symons, 1967) isolated in the pilot samples during magnetisation, with larger symbols represing directions with a stability index greater than 2.5, i.e. high stability. Stereographic plots (d) and (e) show changes in direction of pilot samples during incremental demagnetisation and the changes in intensity are shown in (f).


(a)
(b)
(c)
(d)
(e)
(f)
R - J
16.1
15.1
5.1
18.1
6.4
19.1
6.2
17.1
14.1
5.1
4.1
9.1
Intensity (Am⁻¹, log scale)
1
10⁻¹
10⁻²
10⁻³
10⁻⁴
19.1
3.1AF
17.1
15.1
16.1
4.1
9.1
6.2
18.1
5.1
14.1
0
200
400
600
Temperature (°C) or Peak Alternating Field (10⁻⁴Wb/m⁻²)

small decrease, while pilot specimen RA 18.1 showed a rapid decrease and was unmeasurable at temperatures above 300°C. These Middle Jurassic pilot specimens also showed different behaviour, with sample RA 17.1 remaining almost constant in direction, RA 18.1 was initially similar in direction to RA 17.1, but then moved well away from it at temperatures over 50°C, and RA 19.1 changed direction, by over 100°, immediately after heating, but then retained this direction throughout all subsequent heating. It is possible that the directional change of RA 18.1 may have indicated a movement away from the present earth's magnetic field towards a stable direction isolated in RA 19.1, but this is not clear and the decrease in the intensity prevents further observation. Pilot specimens from the Lower Jurassic sites (RA 14—16) showed similar behaviour to each other with the decrease in intensity being more marked in RA 14.1 than in the other two. The directional change in pilot specimens RA 14.1 and 15.1 was similar, both remaining closely grouped about the same direction, which is similar to that of RA 17.2, but RA 16.1 showed small, but possibly systematic, changes in direction over different temperature ranges. On the basis of a statistical analysis of the directional properties (Table II), all remaining specimens were heated to 300°C. At this temperature, five of the six sites had internal scatters of less than 20° (c.s.d.) and three of the sites were grouped close to the present geomagnetic field (RA 15—17) and the site direction of one site (RA 19) was approximately antiparallel to the present geomagnetic field. The effect of correction for bedding tilt was small, with a tendency to slightly increase the between site scatter, especially before demagnetisation, but the change (6°) and the number of sites were too small to consider that there was any distinct evidence for the magnetisation having been acquired before or after tilting. Site RA 16, after partial demagnetisation, is well away from the other Jurassic sites and the analysis for the mean Jurassic direction has been made with and without the inclusion of this site, although there are no obvious grounds for rejecting this site.

Jurassic—Cretaceous laccolith (Fig. 3)

All three sites (RA 10—12) were tightly grouped both before and after partial demagnetisation, with all three becoming more shallow after treatment (Table I). The three pilot specimens all showed good stability (Table II). As all sites were several metres apart from each other, it is considered that these three sites give a good average direction of magnetisation for the time that the laccolith was intruded.

Cretaceous (Fig. 3)

Only the three Upper Cretaceous sites (JE1, 9 and 10) had a low internal scatter before partial demagnetisation, and two of these sites were very weakly magnetised (JE1 and 10). Alternating magnetic field partial demag-

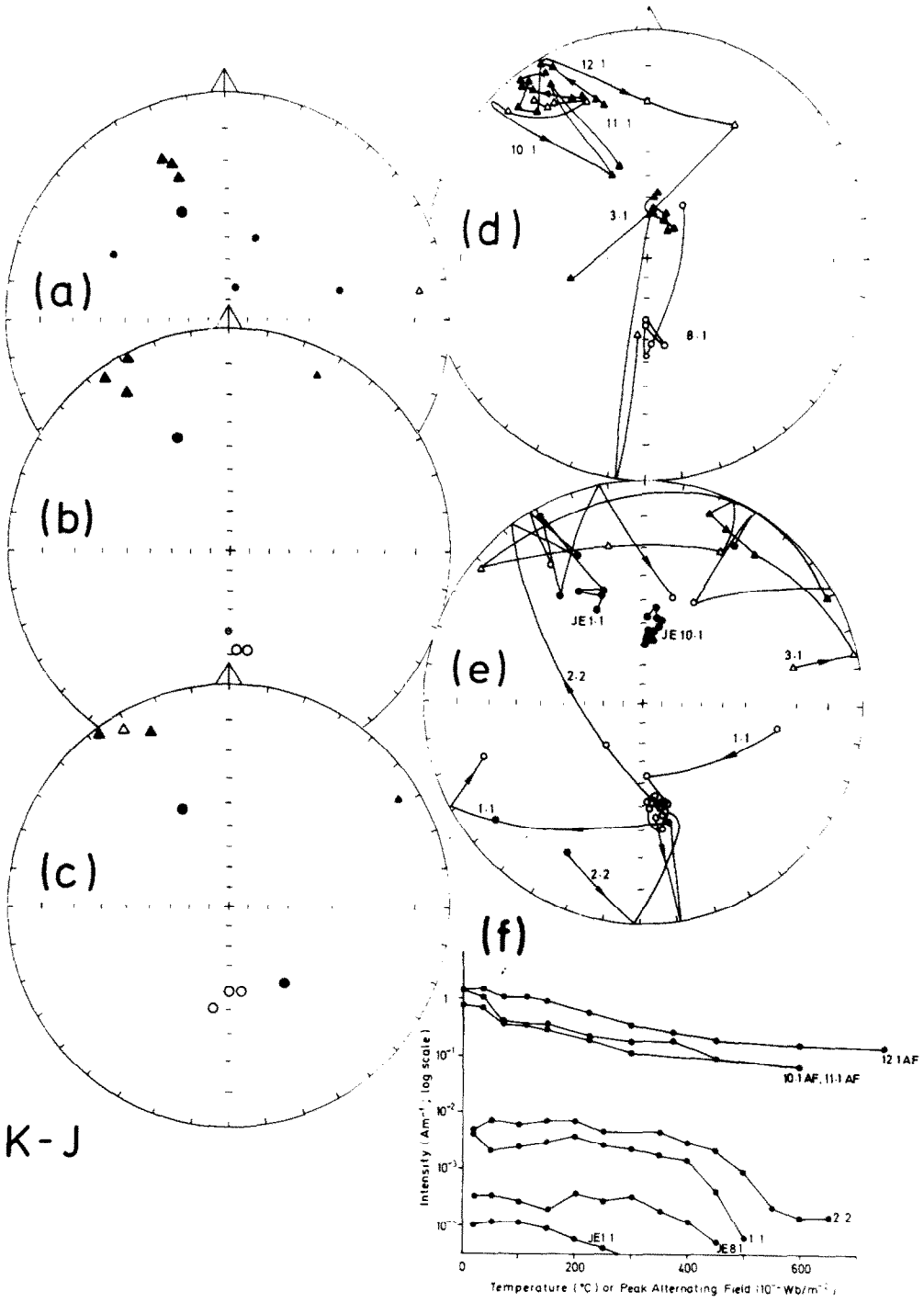


Fig. 3. Jurassic—Cretaceous directions and intensity and magnetisation. The directions and intensitions of remanence are shown as in Fig. 2, but all samples are of sediments and demagnetisation is entirely by thermal means.

netisation of the Lower Cretaceous basalt pilot specimen (RA 3.1) showed the remanence was unstable and, after partial demagnetisation, the directions within this site remained scattered (Tables I and II). The Senonian sites (JE8) showed poor stability and JE10 became too weakly magnetised for adequate analysis of its stability. However, both sites showed a good internal consistency after partial demagnetisation, although with distinctly different mean directions; all five sedimentary sites, in fact, showed good internal consistency after partial demagnetisation.

Mean site values

As only three Triassic sites show low internal scatter after partial demagnetisation and all three site mean directions are somewhat scattered, it is thought that the overall mean direction of these sites can only be considered to indicate, but not define, the Traissic geomagnetic field in Israel. However, the mean pole is clearly in much closer agreement with the mean African Triassic than with the European pole (Fig. 4). Similarly, the Israeli Jurassic data are somewhat scattered, although there is also a distinctly better agreement with the African mean Jurassic poles rather than with the European. The laccolith site directions show high internal and between-site precision, with the mean pole close to the Triassic Israeli pole and suggests that the age is more likely to be Jurassic than Cretaceous.

Comparison of the Israeli results with those for Europe is hampered by the lack of reliable data from those parts of Europe that have not been affected by the Alpine orogeny, particularly for Jurassic and Cretaceous times. On this basis, the mean European data for these periods are used but are supplemented by rotating the North American polar wandering curves relative to Europe (Van der Voo and French, 1974) using sea-floor magnetic anomalies and geometric constraints for the corresponding periods (Fig. 4). However, such a procedure is hampered by uncertainties about the reliability of such rotations and also by the degree of scatter in North American Jurassic poles. Nonetheless, it is clear that the agreement between the Israeli and extended "European" poles is poor for Triassic times and not very satisfactory for either Cretaceous or Jurassic times. As the Israeli data all come from west of the Dead Sea Fault, it seems unreasonable to rotate the Israeli data with Arabia, although such a rotation would marginally improve the agreement with the "European" pole positions.

The African and Israeli pole positions can be compared directly as both lie on the same plate (Fig. 4). The Israeli Triassic and Essexite poles lie outside the small circle of confidence for the African Triassic—Jurassic, but are clearly much nearer the African than "European" positions. The Israeli Jurassic and Cretaceous poles lie within the circles of confidence of the corresponding African poles, but are only slightly nearer to the African than "European" pole positions. Nonetheless, considering all of the Israeli data, there is clearly a better agreement with African than postulated European pole positions, confirming the extension of the African plate as far north as Jerusalem.

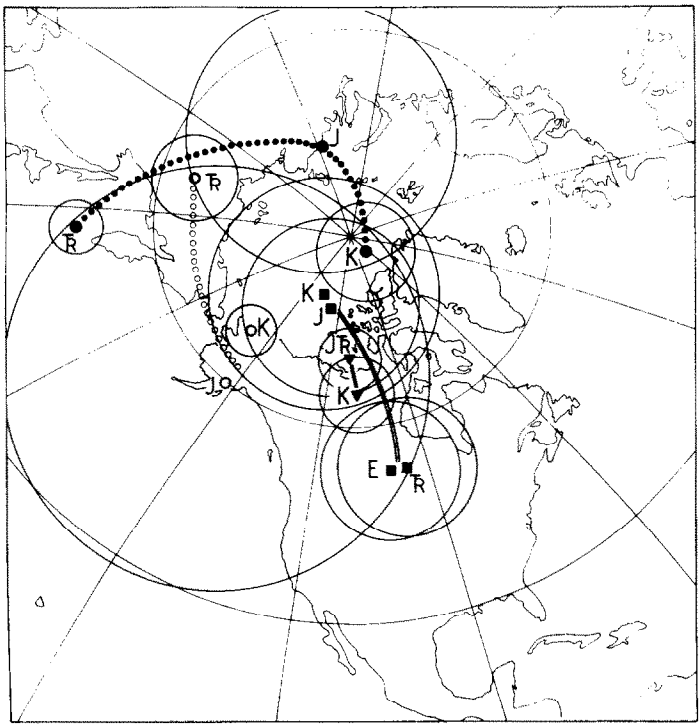


Fig. 4. Pole positions for selected age groups. The mean pole positions and circles of 95% confidence are shown for Israeli (squares), African (triangles) and true European (dots) and postulated European (circles) data, giving equal weight to formation means for the continental regions and equal weight to each site for the Israeli observations. The age groups are for Triassic ($\bar{R}$), Jurassic (J), Cretaceous (K) and the Israeli Jurassic—Cretaceous Essexite laccolith (E). The postulated European polar path has been determined by rotating North American data to possible palaeogeographic positions for these periods making use of oceanic magnetic anomalies. The "true" European data are after Krs (1978) but it is emphasised that there appear to be no completely reliable Jurassic or Cretaceous data for extra-Alpine Europe and that the Jurassic data for the rotated North American path are widely scattered.

DISCUSSION

Although the palaeomagnetic results are still preliminary, there seems little doubt that the Triassic to Cretaceous poles measured in Israel are much closer to those for Africa than those for Europe, and far away from the Lebanese pole determinations. The small (10°−15°) counterclockwise deviation of the Israeli poles from the African ones, although not satisfactorily statistically confirmed, may correspond to the rotation arrived at by geological considerations, namely 6° counterclockwise rotation of Israel together with Arabia away from Africa associated with the opening of the Red Sea,

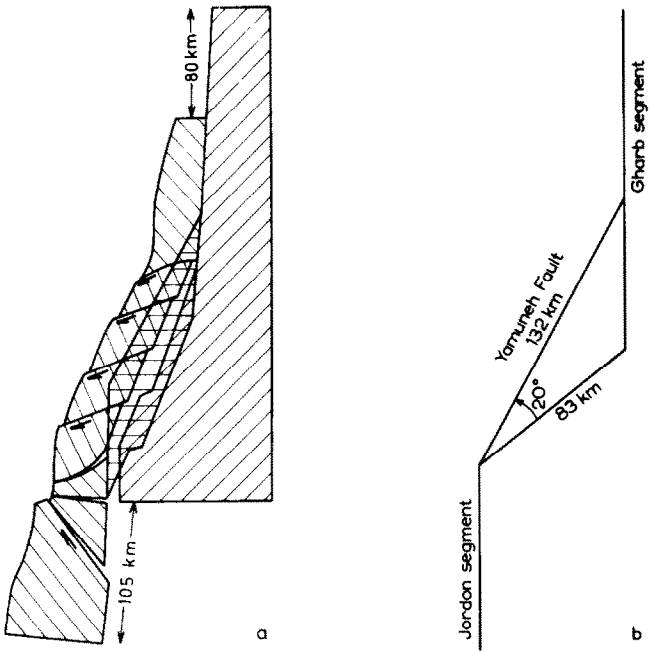


Fig. 5. A model for the tectonics of the Levantine region.

and an additional 7° rotation in the same direction indicated by the internal deformation of southern Israel and northern Sinai, where long east—west strike-slip faults (perpendicular to the Dead Sea Rift), spaced 20—40 km apart, displace dykes of Miocene age and various structures by about 2—4 km each (Bartov, 1973). In any case, the paleomagnetic measurements leave little doubt that a wide angle exists between the Israeli poles and the Lebanese poles (Van Dongen et al., 1967; Gregor et al., 1974), yet the continuity of geological structures between Jerusalem and Beirut shows clearly that there was no plate (or microplate) boundary between the two. This cannot, therefore, account for the large rotation recorded by the palaeomagnetic results. It has been suggested (Freund, 1970; Freund et al., 1970) that a certain part of the transversal shortening of Lebanon, which is caused by the 105 km south—north left-lateral shear along the Dead Sea rift colliding against the N30°E trending Yamuneh segment of this rift (Fig. 5), is taken up by the counterclockwise rotation of faulted blocks of Lebanon. Mt. Lebanon is dissected by numerous E—W to WSW—ENE vertical faults with rather moderate vertical displacements (Dubertret, 1954). Renouard (1955) suggested that right-hand strike—slip movements occur along these faults, and in one case this has been confirmed in northern Galilee (Freund, 1970). Clearly right-hand strike—slip movements on east—west faults within a south—north left-hand shear zone is indicative of counterclockwise rotation of the blocks between the faults (as illustrated on Fig. 5). As the amount of

strike–slip movements on the various faults in Lebanon has not been measured, only a modest amount of displacement is assumed (represented as 3 km on each of five faults) leading to a small rotation (less than 10°) and moderate transversal shortening and longitudinal extension (10%).

The 30° difference between the Jurassic poles and the Cretaceous poles of Lebanon recorded in both studies of this country (Van Dongen et al., 1967; Gregor et al., 1974) seems, however, to contradict the above suggestions as the deformation of Lebanon, related to the Dead Sea Rift, is later (since Early Tertiary or Late Cretaceous at the earliest). This 30° difference cannot be related to the movements of Africa between Jurassic and Cretaceous as the poles of the two periods of both Africa and Israel are too close together. Nor is it likely that the Early Mesozoic tectonic phase in the Levant, reflected by the volcanic rocks in the area marked by extensional features associated with the formation of the Eastern Mediterranean basin (Freund et al., 1975), is responsible for this rotation as this phase did not lead to similar rotations in Israel. (However, the present data are still too sparse to reject this possibility.) However, it is suggested here that different grouping of the available palaeomagnetic data shows that the deformation can be related to the specific movements of individual Lebanese fault blocks.

When the location of sampling sites in the Lebanon is examined it appears that the results form distinct groups according to their location, rather than to their precise age. For example the Cretaceous results from Beskinta (D = 112.6°), Dara (D = 97.4°) and Col of Zahle (D = 63.1°), which are located east and northeast of Beirut in the vicinity of Jurassic sites, showing declinations about 100°, are rather close to the latter than to the Cretaceous sites farther south (Abey, D = 125°) or north (Ehmej, Laklouk, D = 136°) and were obtained from areas dissected by numerous dense faults, oriented northeast, and hence prone to large shear and rotation than the areas where a smaller amount of faulting is observed (e.g. Ehmej, Laklouk), or where faults oriented west-northwest (Abey) may indicate smaller rotation. Indeed, the deformation of Lebanon would not be expected to be homogeneous. Close to faults much larger rotations of both Jurassic and Cretaceous rocks have occurred, leading to declinations of 63°. This shows that the palaeomagnetic results of the Lebanese rocks may indeed record the structural deformation of this country.

The relationships between the rotations of faulted blocks (r), their original deviation from the direction of shortening (s), the ratio of the displacement along the faults (d) to the width of the faulted blocks (w) and the amount of extension ($\sqrt{\lambda_e}$) is given by the following (Freund, 1974):

$$d/w = \sin r/\cos s \cos(s + r) \tag{1}$$

$$\sqrt{\lambda_e} = 1/\sqrt{\lambda_s} = \cos s/\cos(s + r) \tag{2}$$

Only a few of these parameters can be derived at present in Lebanon. The angle ($s + r$) is that between the faults and the direction normal to Yamuneh

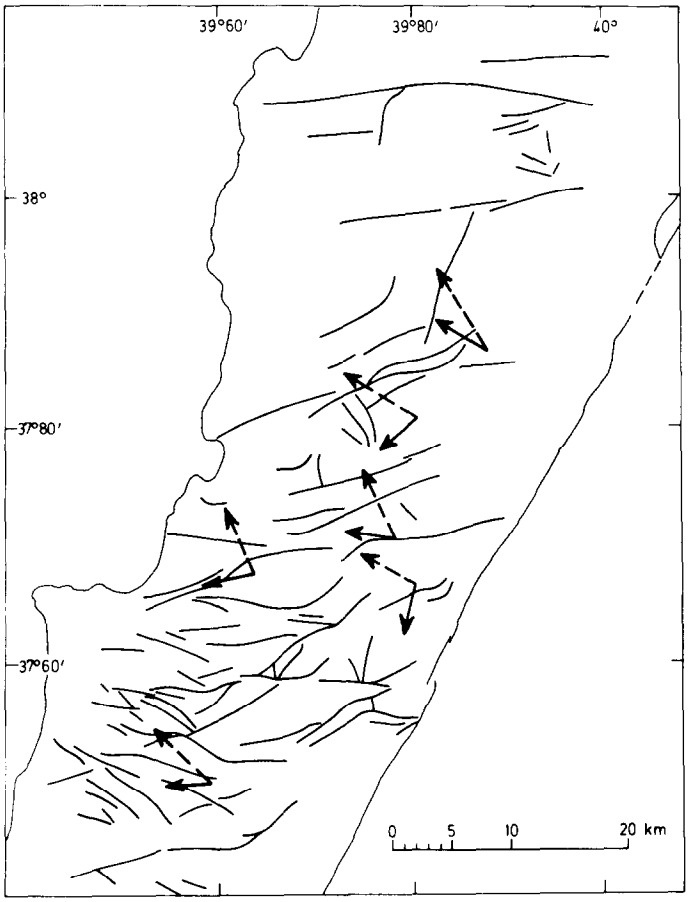


Fig. 6. The fault system in the Lebanon area. The dashed arrows indicate directions of cross bedding as observed today (Kana'an, 1966) and the solid arrows indicate the direction of cross bedding when corrected for the palaeomagnetically determined rotation of individual fault blocks. The fault system is after Dubertret (1955).

fault. This angle varies from 30° near Laklouk and to 55° east of Beirut (Fig. 6). This 25° difference agrees well with the 28° difference between the palaeomagnetic measurements at these two localities (Fig. 6). The original (prerotational) direction of these faults can be estimated to be half the minimum angle between sets of faults which appear as conjugate sets on the geological map. As several such occurrences show an angle of about 40°, the angle s can be estimated as some 20°. However, s can also be calculated from the horizontal displacement on faults, shown by Renouard (1955, Fig. 10) in Ehmej—Laklouk area. According to this map the dextral displacements are about 3 km, on faults lying 12 km apart. Accordingly, the ratio d/w is 0.25, and as $(s + r)$ in this region is 30°, r is found by (1) to be 16°, and $s = 14°$.

If this calculation is correct, the local rotation of the fault blocks in this place is 16°. With the additional 6° rotation of the Arabian block (the opening of the Red Sea) the palaeomagnetic poles of this site should be located 22° anti-clockwise from the African pole. In fact they are 40° away (Gregor et al., 1974), so that some 18° are not accounted for by the present structural considerations.

Checking this result for the area east of Beirut, r would be $[(s + r - s] = 55° - 14 = 41°$, which with the 6° rotation of the Arabian block would amount to 47°. Again 21° of the 68° palaeomagnetic rotation are not accounted for. The d/w ratio at Beirut area should amount to 1.2 according to (1), and since the faults there are on the average 4 km apart, the dextral movement along them should amount to 5 km. The geological map of Lebanon (Dubertret, 1954) shows a 5 km offset of a basaltic body on the fault near Bikfaya, but whether this offset is a real dextral displacement remains to be seen by closer geological study. Utilizing these results with (2), the SW—NE extension at Laklouk—Ehmej amounts to 1.14, and that at Beirut to 1.69.

This amount of extension can be checked against the model, shown diagrammatically in Fig. 5 (Freund et al., 1970), as follows. The 105 km movement along the Dead Sea Rift would cause an overlap of 6720 km^2, accounting for an excess volume of 168,000 km^3 in a crust of 25 km (according to Lort, 1973). A certain part of this excess material is probably stored in the high mountains of Lebanon and Anti Lebanon (and in their roots). This part can be estimated to be 90,000 km^3 since the area above 1000 m is 9000 km^2 and their average excess height is 1500 m with the excess thickness of the mountains and their roots should be about 10 km. The remainder (about 78,000 km^3) must be accommodated in the lengthening of these mountains and the Yamuneh fault between them by about 49 km, to their present length of 132 km (Fig. 5b). This is an extension of 1.59, which is in line with the extension derived from the rotation of the faulted blocks. At the same time this extension requires a net 20° anti-clockwise rotation of the entire Lebanon.

This rotation is almost exactly that which remained unaccounted for by the internal rotation of the faulted blocks in Lebanon combined with the rotation of the entire Arabian block.

A rotation of 20° of Lebanon as a whole requires the stretching of eastern Galilee relative to western Galilee in the way shown diagrammatically in Fig. 5. It has been shown previously (Freund, 1970) that the measured deformation of the large features of a part of this area (Galilee) shows a 4° bending to the northwest, in line with this result, but still far too small. However, if the entire area showing this type of deformation is considered, and if smaller deformations (joints, minor faults, strain of the rocks) go in the same direction, this gap may be overcome.

It is desirable that field studies are undertaken to examine the faults, displacement, rotations etc., but a minor attempt in this direction can

already be attempted as Gregor et al. (1974) quote eighteen crossbedding directions measured in the Gres du Liban by Kana'an (1966). Six of these measurements are located close to the palaeomagnetic sites, and they are rotated so that the latter reach declination 165° (Fig. 6). As a consequence of this rotation all the crossbeddings rotate into the northwestern quadrant between N25° and N60°W, in agreement with the original direction (Karcz, 1965). Although little weight can be placed on such few observations, there is clearly some evidence in support of the proposed rotation of a series of fault blocks as a more realistic interpretation for the Cenozoic history of this area. In fact, the palaeomagnetic measurements clearly determine the total rotation, whereas each tectonic strain measurement shows only a small component of the rotation. Certainly, much detailed work has to be carried out in Israel and Lebanon to substantiate the fragmentary data and result presented here, and to give a full account of the deformations and rotations in northern Israel and Lebanon. Nonetheless, the existing data permit the suggestion for the explanation of the palaeomagnetic data from the Jurassic basalts of the Lebanon by a summation of a ca. 20° rotation of Africa with a 6° rotation of Arabia, an approximate 20° bending of whole Lebanon to northwest and a ca. 41° internal rotation on small faulted blocks separated by dextral displacements. It is also relevant that the site of the individual tectonic units postulated in the Lebanon is similar to that determined for units in the Aegean, using sea-level change observations (Fleming, 1978), and for California based on palaeomagnetic studies (Kammerling and Luyendyk, in press). Clearly much more detailed palaeomagnetic and tectonic studies are required to establish the reliability of this proposed model, but such a model certainly appears a more realistic interpretation of the Levantine area than a model involving only one or two microplates. Furthermore, if this interpretation is correct, it indicated the possibility of measuring very localised tectonic strains by palaeomagnetic methods.

REFERENCES

Bartov, J., 1973. A Structural and Palaeogeographical Study of the Central Sinai Faults and Domes. Thesis, Hebrew University, Jerusalem, 143 pp. (in Hebrew).

Dubertret, L., 1954. Carte geologique au 200,000. Feuille de Beyrouth, avec notice explicative. République Libanaise.

Flemming, N.C., 1978. Holocene eustatic changes and coastal tectonics in the northwest Mediterranean: implications for models of crustal consumption. Philos. Trans. R. Soc., London, Ser. A, 289: 405—458.

Freund, R., 1970. The geometry of faulting in the Galilee. Isr. J. Earth Sci., 19: 117—140.

Freund, R., 1974. Kinematics of transform and transcurrent faults. Tectonophysics, 21: 93—134.

Freund, R. and Tarling, D.H., in prep. Magnetic studies of samples from the mottled zone of the Hazeva Formation.

Freund, R., Garfunkel, Z., Zak, I., Goldberg, M., Weissbrod, T. and Derin, B., 1970. The shear along the Dead Sea Rift. Philos. Trans. R. Soc. London, Ser. A, 267: 107—130.

Freund, R., Goldberg, M., Weissbrod, T., Druckman, Y. and Derin, B., 1975. The Triassic Jurassic structure of Israel and its relation to the origin of the Eastern Mediterranean. Isr., Geol. Surv., Bull., 65.

Gregor, C.B., Mertzman, S., Nairn, A.E.M. and Negendank, J., 1974. The palaeomagnetism of some Mesozoic and Cenozoic volcanic rocks from the Lebanon. Tectonophysics, 21: 375—395.

Helsley, C.E. and Nur, A., 1970. The palaeomagnetics of Cretaceous rocks from Israel. Earth Planet. Sci. Lett., 8: 403—410.

Kammerling, M.J. and Luyendyk, in press. Tectonic rotation of the Santa Monica Mountains region, western transverse ranges, California, suggested by palaeomagnetic vectors. Geology.

Kana'an, F., 1966. Sedimentary Structures and Facies Variations in the Basal Cretaceous Sandstones of Central Lebanon. Thesis, American University, Beirut, 117 pp.

Karcz, Y., 1965. Lower Cretaceous fluviatiles in the Levant. Nature, 207: 1145—1146.

Krs, M., 1978. Implication of statistical evaluation of Phanerozoic palaeomagnetic data (Eurasia, Africa). Abstr. Eur. Geophys. Soc., PM32.

Lort, J.M., Limond, W.Q. and Gray, F., 1974. Preliminary seismic studies in the Eastern Mediterranean. Earth Planet. Sci. Lett., 21: 355—366.

Renouard, G., 1955. Oil prospects of Lebanon. Bull., Am. Assoc. Pet. Geol., 29: 2125—2169.

Tarling, D.H. and Symons, D.T.A., 1967. A stability index of remanence in palaeomagnetism. Geophys. J.R. Astron. Soc., 12: 443—448.

Van der Voo, R. and French, R.D., 1974. Apparent polar wandering for the Atlantic bordering continents. Late Carboniferous to Eocene. Earth Sci. Rev., 10: 99—119.

Van Dongen, R.G., Van der Voo, A. and Raven, Th., 1967. Palaeomagnetic research in the central Lebanon mountains and in the Tartous area (Syria). Tectonophysics, 4: 35—53.